

Institute of Actuaries of India

Subject CT1 – Financial Mathematics

December 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) The return achieved over the period up to the maturity date of the bond will be uncertain for various reasons:

- i. Reinvestment Risk - The investor may need to reinvest the coupon payments. The terms that will be available for reinvestment are not known at outset.
- ii. Early redemption - For an investor who plans to sell before redemption, the ultimate sale price is not known at outset.
- iii. Real return - The real return (i.e. in excess of inflation) is uncertain. If inflation turns out to be higher than expected at outset, the real returns from fixed interest bonds will be lower than originally anticipated.
- iv. Tax rates - Tax rates may change, affecting the income and capital proceeds received by the investor.

[4]

- ii) The investors in the swap agreement are exposed to the following risks:

- i. Market risk - Market conditions may change resulting in increase in the present value of the net outgo under the agreement.
- ii. Credit risk - Risk that the counterparty may default on its payments.

[2]

iii)

- a) A has bought a call option and being the buyer, has the right to exercise the option and in that case, counterparty (B in this case) will be obliged to complete the transaction. A is likely to choose to buy the underlying asset if the market price exceeds the exercise price i.e. $S_t > K$.

Similarly, C has bought a put option and being the buyer, has the right to exercise the option and A will be obliged to complete the transaction. C is not likely to choose to sell the underlying asset if the market price exceeds the exercise price i.e. $S_t > K$.

Hence in this scenario, A will exercise the option and B will be obliged to complete the transaction. And, C will not exercise the option.

[1.5]

- b) A is not likely to choose to buy the underlying asset at strike price if the market price falls below the exercise price i.e. $S_t < K$.

Similarly, C is likely to choose to sell the underlying asset if the market price exceeds the exercise price i.e. $S_t < K$.

[1.5]

- iv) $A = 78.94 \times 1000 e^{0.035 \times 8/12}$

$$B = \frac{1}{2} \times 0.07 \times 100,000 e^{0.035 \times (8/12 - 5/12)}$$

$$C = A - B = \text{INR } 77,273$$

The redemption rate of 114% is not required, only the price.

[4]

[13 Marks]

Solution 2:

- i) The equation of value equates the value of the payments made to the value of the payments received at an appropriate rate of interest. The values can be determined at any point in time, but it is common to equate present values, i.e. the values at time 0 [1]

$$\text{ii) } \left(1 - \frac{91}{365} \times 0.07\right) = \left(1 + \frac{i^{(2)}}{2}\right)^{\frac{-91}{182.5}}$$

$$\Rightarrow i^{(2)} = (0.982548^{\frac{-182.5}{91}} - 1) \times 2 = 0.03594 \times 2$$

$$\Rightarrow i^{(2)} = 0.07188 = \mathbf{7.19\% \text{ p.a. convertible half-yearly}} \quad [3]$$

- iii) Cost of vacation in today's time = Rs. 50,00,000
 Cost of vacation in 10 years' time = $50,00,000 \times (1.05)^{10} = 81,44,473$

Therefore, after 10 years, Yash needs Rs. 81,44,473 for which he has to contribute every month which will grow at 8.15% p.a. effective.

Let X be the monthly savings.

$$12X s_{\overline{10}|}^{(12)} = 81,44,473 \text{ @ } 0.0815$$

$$s_{\overline{10}|}^{(12)} = 15.1275$$

$$\Rightarrow X = \frac{81,44,473}{12 \times 15.1275} = 44,865.71$$

Hence, Yash should save a nominal amount of Rs. 44,866 from his salary every month.

[3]

[7 Marks]

Solution 3:

- i) $a^{(p)}_{n|}$ is the value at time 0 of a level annuity payable p thly in arrear at the rate of 1 per unit time over the time interval $(0, n)$.

For this annuity the payments are made at time $1/p, 2/p, 3/p, \dots, n$ and the amount of each payment is $1/p$.

[1]

ii) Algebraically:

$$\begin{aligned}
 a^{(p)}_{n|} &= \sum_{t=1}^{np} \frac{1}{p} v^{t/p} \\
 &= \frac{1}{p} v^{1/p} \frac{(1 - v^n)}{1 - v^{1/p}} \\
 &= \frac{1 - v^n}{p((1+i)^{1/p} - 1)} \\
 &= \frac{1 - v^n}{i^{(p)}} \\
 &= \frac{i}{i^{(p)}} \frac{(1 - v^n)}{i} \\
 &= \frac{i}{i^{(p)}} a_{n|}
 \end{aligned}$$

General reasoning:

By definition, a series of p payments, each of amount $i^{(p)}/p$, in arrear at p thly subintervals over any unit time interval, has the same value as a single payment of amount i at the end of the interval. By proportion, p payments, each of amount $1/p$ in arrear at p thly subintervals over any unit time interval, have the same value as a single payment of amount $i/i^{(p)}$ at the end of the interval. -----(1)

Consider now the annuity for which the present value is $a^{(p)}_{n|}$. Using (1) above, the p payments after time $r-1$ and not later than time r have the same value as a single payment of amount $i/i^{(p)}$ at time r . This is true for $r = 1, 2, \dots, n$, so the annuity has the same value as a series of n payments, each of amount $i/i^{(p)}$ at times $1, 2, \dots, n$. This means that: $a^{(p)}_{n|} = i/i^{(p)} a_{n|}$

[4]

iii) $(\bar{I}\bar{S})_{10|} @ 11\%$ is the accumulated value, at the end of 10 years, of an increasing annuity which has a constant rate of payment r throughout the r^{th} year ; $1 \leq r \leq 10$. The annuity is payable continuously for a period of 10 years at a force of interest equivalent to an interest rate of 11% p.a.

$$(\bar{I}\bar{S})_{10|} @ 11\% = (\bar{I}\bar{a})_{10|} @ 11\% \times (1+i)^{10} = \frac{\ddot{a}_{10|} - 10v^{10}}{\delta} \times (1+i)^{10} = 28.89248 \times 1.11^{10} = 82.0379$$

$$\ddot{a}_{10|} = \frac{1 - v^{10}}{d} = 6.537059$$

$$\delta = \ln(1 + i) = \ln 1.11 = 0.104360$$

$$d = iv = 0.099099$$

[3]

iv) Accumulated value of Rs. 1 invested at time 0 after 5 years for each of the options is:

Option A:

$$\begin{aligned} \text{Accumulated Value} &= 1 * 1 / [(1 - 0.07/2)^4] * (1 + 0.08/4)^{12} \\ &= 1 * (1 - 0.07/2)^{-4} * (1 + 0.08/4)^{12} \\ &= 1 * 1.1532 * 1.2682 \\ &= 1.46248 \sim 1.4625 \quad \text{----- (1)} \end{aligned}$$

Option B:

$$\begin{aligned} A(0,5) &= \exp \int_0^5 [0.05 + 0.003t] dt \\ &= \exp [0.05t + 0.003 \frac{t^2}{2}]_0^5 \\ &= \exp [0.05 * (5) + 0.003 * \frac{(5)^2}{2}] \\ &= \exp [0.25 + 0.0375] \\ &= \exp [0.2875] \\ &= 1.333091 \end{aligned}$$

$$\text{Accumulated Value} = 1 * 1.333091 \sim 1.3331 \quad \text{----- (2)}$$

From (1) and (2) above, the maximum accumulated value of the investments after a period of 5 years is under Option A. Therefore, to maximise the return, the investor should invest 75% in Option A and 25% in Option B.

Hence, the accumulated amount of Rs. 10,000 after 5 years is :

$$1000 [0.75 * 1.4625 + 0.25 * 1.3331]$$

$$= \text{Rs. 14301.5}$$

[4]

[12 Marks]

Solution 4:

i) Let's assume monthly instalment is Rs. X.

$$\text{PV of loan taken} = \text{PV of monthly instalments}$$

$$2 * 12,50,000 * \ddot{a}_{27}^{(2)} @ 10\% = 12 X * a_{107}^{(12)} @ 10\% * v^2 \quad \text{----- (1)}$$

We know that,

$$\ddot{a}_{n1}^{(p)} = i/d^{(p)} a_{n1} \quad \text{and} \quad a_{n1}^{(p)} = i/i^{(p)} a_{n1}$$

From table, values at 10% p.a. interest rate are

$$a_{27} = 1.7355, a_{107} = 6.1446, d^{(2)} = 0.093075, i^{(12)} = 0.09569$$

$$\ddot{a}_{27}^{(2)} = \frac{0.10 * 1.7355}{0.093075} = 1.864625$$

$$a_{107}^{(12)} = \frac{0.10 * 6.1446}{0.09569} = 6.421361$$

$$v^2 = (1+i)^{-2} = (1+0.1)^{-2} = 0.826446$$

Substituting the values in equation (1) above

$$2 * 12,50,000 * 1.864625 = 12 X * 6.421361 * 0.826446$$

$$4661562.50 = 63.682897 X$$

$$X = 4661562.50 / 63.682897$$

$$= 73199.60 \sim 73200$$

Hence, monthly instalment will be Rs. 73,200.

[4]

ii) Interest portion in first instalment will be

= Loan outstanding at the end of 2nd year * interest for one month (as instalments are monthly)

$$= 2 * 12,50,000 * S_{27}^{(2)} * i^{(12)} / 12$$

$$= 25,00,000 * \ddot{a}_{27}^{(2)} * (1+i)^2 * i^{(12)} / 12$$

From (i) above, $\ddot{a}_{27}^{(2)} = 1.864625$ and from tables, $i^{(12)} = 0.09569$

$$\begin{aligned} \text{Interest in first instalment} &= 25,00,000 * 1.864625 * (1.10)^2 * 0.09569 / 12 \\ &= 44978.21 \end{aligned}$$

Hence, interest portion in first instalment is Rs. 44978.21

[2]

iii) Outstanding loan after paying instalment for 5 years is

= Present value of outstanding instalments at that time

= $12 * \text{Instalment} * a_{n-t}^{(12)}$, where $n-t = 10-5 = 5$ yrs

= $12 * 73,200 * a_{57}^{(12)}$

From table $a_{57} = 3.7908$, $i^{(12)} = 0.09569$

= $12 * 73200 * 3.7908 * 0.1 / 0.09569$

= 3479818.92

Hence, outstanding loan after paying EMI for 5 years will be Rs. 34,79,818.92

Let us assume EMI for revised outstanding duration of 2 years is Rs. Y

So, outstanding loan after paying 5 years instalment = $12 * Y * a_{27}^{(12)} @ 10\%$

Therefore,

$3479818.92 = 12 * Y * i/i^{(12)} a_{27}$

$3479818.92 = 12 * Y * 0.10 * 1.7355 / 0.09569$

$3479818.92 = 21.76403 * Y$

$Y = 159888.54$

Therefore, revised instalment will be Rs. 1,59,888.54

[5]

iv) Amount of interest paid after reschedule

= Total amount paid in instalment – outstanding loan after paying instalment for 5 years (before reschedule)

= $24 * 159888.54 - 3479818.92$

= 357506.04

Therefore, total interest after reschedule is Rs. 3,57,506.04

[2]

[13 Marks]

Solution 5:

i)

- a) Discounted Payback Period of a project: An investment project usually requires expenditure (or investment) in the early years before returns (or profits) emerge in the later years. So during the early years the 'project account' will be overdrawn and interest will be incurred at the borrowing rate.

The discounted payback period is the time from the beginning of the project until the 'project account' balance first becomes positive

Or

Discounted payback period ends at the first point in time such that the net present value (or accumulated value) of project cashflows up to that time is positive.

[2]

- b) The internal rate of return for an investment project is the effective rate of interest that equates the present value of income and outgo, i.e. it makes the net present value of the cashflows equal to zero.

If all the payments for the project were transacted through a bank account that earned interest at the same rate as the internal rate of return, the net proceeds at the end of the project (i.e. the accumulated profit) would be zero. A higher internal rate of return indicates a more "profitable" project.

[2]

- ii) Working in terms of monthly units of time, effective monthly interest rate =
 $i(12)/12 = 0.086488/12$
 $= 0.007207333$

The project plan is as below:

<u>Cashflow item</u>	<u>Timing</u>	<u>Amount (in crores)</u>
Cost of construction	Every month starting from Time 0	- $[10 - t(0.5)]$ where $t = 1, 2, \dots, 19$ (in months)
Maintenance Cost of 24 months	Every month, starting Immediately after completion	- 0.10
Rental income	At the start of each month beginning with 21st month	+ 2.1

Let C be the present value of cost of construction.

Therefore, $C = 10 + (10 - 0.5)v + [10 - (0.5)2]v^2 + \dots + [10 - (0.5)19]v^{19}$

$$= 10 \ddot{a}_{20}| - 0.5 (1a)_{19}|$$

$$= 10 \frac{1-v^{20}}{d} - 0.5 \frac{(\ddot{a}_{19}| - 19v^{19})}{i} = 186.97044 - 86.57$$

$$= 100.3910$$

Hence, the present value of cost of construction is Rs. 100.3910 crores.

$$v^{19} = 0.872452$$

$$v^{20} = 0.866208$$

$$i = 0.007207$$

$$d = 0.007156$$

$$\ddot{a}_{20}| = 18.697044$$

$$\ddot{a}_{19}| = 17.824592$$

[5]

iii) Let t be the DPP in months

$$\Rightarrow -C + 2.1 * \ddot{a}_{t-20}| v^{20} - 0.1 * \ddot{a}_{t-24}| v^{24} \geq 0$$

$$= -100.3910 + 2.1 * \ddot{a}_{4}| v^{20} + 2.1 * \ddot{a}_{t-24}| v^{24} - 0.1 * \ddot{a}_{t-24}| v^{24}$$

$$= -100.3910 + 2.1 * 3.957272 * 0.866214 + 2 * \ddot{a}_{t-24}| v^{24}$$

$$= -100.3910 + 7.198473 + 2 * 0.841686 * \ddot{a}_{t-24}|$$

$$= -93.192527 + 1.683372 * \ddot{a}_{t-24}|$$

$$= -93.192527 + 1.683372 * \frac{(1-v^{t-24})}{d}$$

$$= -93.192527 + 1.683372 * \frac{(1-v^{t-24})}{0.007155}$$

$$= -93.192527 + 235.2721174 * (1 - v^{t-24})$$

$$93.192527 = 235.2721174 * (1 - v^{t-24})$$

$$93.192527 = 235.2721174 - 235.2721174 * v^{t-24}$$

$$93.192527 - 235.2721174 = 235.2721174 * v^{t-24}$$

$$142.07959 = 235.2721174 * v^{t-24}$$

$$v^{t-24} = \frac{142.07959}{235.272117}$$

$$v^{t-24} = 0.603895$$

$$\frac{v^t}{v^{24}} = 0.603895$$

$$\frac{v^t}{0.841686} = 0.603895$$

$$v^t = 0.508290$$

$$\frac{1}{(1+i)^t} = 0.508290$$

$$\frac{1}{(1.007207)^t} = 0.508290$$

At t = 100, LHS = 0.48767

t = 90, LHS = 0.523979

Using interpolation t = 95 months

[8]

iv) Let P be the price at which Mall is sold after 15 years i.e. 180 months

Using the calculations in 5iii) above with t = 180 months, the equation of value becomes:

$$142.07959 = 235.2721 v^{(180-24)} + P v^{180}$$

$$P v^{180} = 142.07959 - 235.2721 v^{156} \quad @0.007207333$$

$$v^{156} = 1.0072^{-156} = 0.3265$$

$$v^{180} = 1.0072^{-180} = 0.2749$$

$$0.2749P = 65.2633$$

$$P = \text{Rs. } 237.41 \text{ crores}$$

[4]

[21 Marks]

Solution 6:

- i) A Redemption yield is the yield obtained by the investor if they hold the bond until redemption. This is sometimes also referred as the yield to redemption.

The Running yield is defined as the ratio of the coupon rate to the price per unit nominal of the stock. This is sometimes also referred as the flat yield. [2]

- ii) The mean and variance of i are 0.08 and 0.1^2 respectively.

This means that the mean and variance of $1+i$ are 1.08 and 0.1^2 respectively.

We can then set up simultaneous equations for mean and variance of the lognormal distribution:

$$1.08 = \exp \left[u + \frac{1}{2} \sigma^2 \right]$$

$$0.1^2 = \exp \left[2u + \sigma^2 \right] (\exp(\sigma^2) - 1)$$

Solving these we get:

$$0.1^2 = 1.08^2 (\exp(\sigma^2) - 1)$$

Which gives $\sigma = 0.092395$ and $u = 0.072693$

From the properties of the log normal distribution:

$$(1+i)^{15} \sim \log N(15xu, 15x\sigma^2) = \log N(1.090389, 0.357844^2)$$

$$\text{We require, } P[100,000 \times (1+i)^{15} > 2,50,000] = P[(1+i)^{15} > 2.5]$$

We can evaluate this as follows:

$$P[\ln(1+i)^{15} > \ln(2.5)] = P[\ln(1+i)^{15} > 0.916291]$$

$$= P \left[Z > \frac{0.916291 - 1.090389}{0.357844} \right]$$

$$= P[Z > -0.48652]$$

$$= 1 - 0.313299$$

$$= 68.67\%$$

[6]

[8 Marks]

Solution 7:

- i) Immunization is a strategy of choosing the asset portfolio such that the surplus (viz. assets minus the liabilities) is immune or protected from any small change in interest rate in future. The conditions for Redington's immunization are summarized as follows:

- I. The value of the assets at the starting rate of interest is equal to the value of the liabilities.
- II. The volatilities of the asset and liability cashflow series are equal.

- III. The convexity of the asset cashflow series is greater than the convexity of the liability cashflow series [3]

ii) **Present Value**

$$100,000 (v^{13} + v^{17}) @8\% = 63,797$$

Duration

$$\frac{100,000 (13 v^{13} + 17 v^{17})}{63,797} = \frac{9,37,465}{63,797} = 14.69$$

Convexity

$$\frac{100,000 (182 v^{15} + 306 v^{19})}{63,797} = \frac{1,28,27,788}{63,797} = 201.07$$

[5]

- iii) Let P and Q denote the maturity values of the 10 year and 20 year zero coupon bonds.

Then the present value of the assets is:

$$Pv^{10} + Qv^{20} @8\%$$

By Redington's first condition, this must equal the PV of the liabilities, which is 63,797

This gives us our first equation for finding P and Q:

$$Pv^{10} + Qv^{20} @8\% = 63,797 \dots\dots\dots(1)$$

Duration of the assets is given by:

$$10 Pv^{10} + 20 Qv^{20} @8\%$$

This gives us our second equation for finding P and Q:

$$10 Pv^{10} + 20 Qv^{20} = 9,37,465 \dots\dots\dots(2)$$

Solving equations (1) and (2), we find that P = 73,073 and Q = 139,595.

Hence amounts to be invested are X = Pv^{10} = Rs. 33,847 and Y = Qv^{20} = Rs. 29,950

[5]

- iv) Since the assets are spread over longer period than that of the liability, the convexity of assets is higher and hence portfolio is immunized (no calculation required)

[2]

[15 Marks]

Solution 8:

- i) Let i_1 be the GRY for Stock A. So, the equation of value is:

$$102 = 110v_1$$
$$i_1 = 7.84\%$$

Let i_2 be the GRY for Stock B. So, the equation of value is:

$$102 = 8v_2 + 113v_2^2$$
$$i_2 = 9.25\%$$

Let i_3 be the GRY for Stock C. So, the equation of value is:

$$102 = 12v_3 + 12v_3^2 + 117v_3^3$$
$$i_3 = 12.63\%$$

[6]

ii) The equation of value for the 1-year spot rate is:

$$102 = 110v_1$$
$$i_1 = 7.84\%$$

The equation of value for the 2-year spot rate is:

$$102 = 8v_1 + 113v_2^2$$
$$i_2 = 9.3\%$$

The equation of value for the 3-year spot rate is:

$$102 = 12v_1 + 12v_2^2 + 117v_3^3$$
$$i_3 = 13.12\%$$

[5]

[11 Marks]
